

MTH00050 – Combinatorial Mathematics

Lecture 1: Sets

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(Tentative) Schedule 2026

No.	Date	Location	Topic
1	13/01 16/01	C24 C24/E302	Introduction, Sets, Proofs
2	20/01 23/01	C24 C24/E302	General counting methods
3	27/01 30/01	C24 C24/E302	Inclusion-exclusion principle
4	03/02 06/02	C24 C24/E302	Recurrence relations
5	03/03 06/03	C24 C24/E302	Generating functions
			Mid-term test

No.	Date	Location	Topic
6	10/03 13/03	C24 C24/E302	Graphs (Part I)
7	17/03 20/03	C24 C24/E302	Graphs (Part II)
8	24/03 27/03	C24 C24/E302	Path problems
9	31/03 03/04	C24 C24/E302	Tree problems
10	07/04 10/04	C24 C24/E302	Network flows (Part I)
11	14/04 17/04	C24 C24/E302	Network flows (Part II)
			Final exam

Course topics

0. Introduction
- 1. Sets**
2. General counting methods for arrangements and selections
3. Inclusion-exclusion principle
4. Recurrence relations
5. Generating functions
6. Combinatorial identities and numbers:
Catalan and Stirling
6. Graphs
7. Path problems
8. Tree problems
9. Network flows

Outline

1. Basic definitions
2. Set Relation
3. Mapping
4. Operations
5. Three basic counting principles
 - Adding principle
 - Multiplication principle
 - Dirichlet's principle

What is a Set?

Definition 1. A finite or infinite collection of objects that share a **common property** is called a **set**.

The objects in a set are called the **elements** of that set.

The elements of a set are **distinct**, and **their order is irrelevant**.

- For short, a set is often simply called a set.
- Note: An extension of the concept of a set is a **multiset**, in which an element is allowed to appear multiple times.

Examples

- The collection of the vowels in the word “probability”. $A = \{p, r, o, b, a, i, l, t, y\}$
- The collection of real numbers that satisfy the equation.
- The collection of two-digit positive integers divisible by 5. $B = \{x \text{ in } \mathbb{N} \mid 5 \mid x\}$
- R is the set of real numbers, Z is the set of integers, and N is the set of natural numbers.
- Depending on convention, N may or may not include 0. To be precise, we can use Z^+ , $Z_{\geq 0}$.

The Membership Relation

- Let A be a set and let x be some object.
- If x is **a member of A** , or x is an element of A , or x belongs to A then we write **$x \in A$** .
- If x is **NOT a member of A** , we write **$x \notin A$** .
- Example: $C = \{a, o, i, e, u, v\}$. $i \in C$ and $z \notin C$.

The Empty Set

- The set **with no elements**.
- Also called the **null** set.
- Notation: $\emptyset$.
- Example: The set of real numbers x that satisfy the equation

$$x^2 + 1 = 0.$$

Finite and Infinite Sets

- A finite set is one which can be counted.
 - Example: The set of two-digit positive integers has 90 elements.
- An infinite set is one which cannot be counted.
 - Example: The set of integer multiples of the number 5.



If a mango can be **divided infinitely**, then its parts would share the **same cardinality as the universe**.



Specifying a Set

- List the elements explicitly, e.g.,

$$C = \{a, o, i\}.$$

- List the elements implicitly, e.g.,

$$K = \{5, 10, 15, 20, \dots\}.$$

- Use set builder notation, e.g.,

$$Q = \left\{ x \mid x = \frac{p}{q}, \text{ where } p \text{ and } q \text{ are integers and } q \neq 0 \right\}.$$

Specifying a Set: avoid using implicit expression

- Set $f(x) = a + bx^{0,5} + cx^2 + dx^3$
- With the given values $f(0) = 5, f(1) = 10, f(2) = 15$, and $f(3) = 20$, we set up the following equations:

- $f(0) = a + b(0)^{0.5} + c(0)^2 + d(0)^3 = 5$
- $f(1) = a + b(1)^{0.5} + c(1)^2 + d(1)^3 = 10$
- $f(2) = a + b(2)^{0.5} + c(2)^2 + d(2)^3 = 15$
- $f(3) = a + b(3)^{0.5} + c(3)^2 + d(3)^3 = 20$

- Then we get:

$$a = 5, b = \frac{30}{18 - 9\sqrt{2} + 2\sqrt{3}}, c = \frac{15(5 - 4\sqrt{2} + \sqrt{3})}{18 - 9\sqrt{2} + 2\sqrt{3}}, d = \frac{5(-3 + 3\sqrt{2} - \sqrt{3})}{18 - 9\sqrt{2} + 2\sqrt{3}}$$

- Therefore, $f(4) \approx 23,479 \neq 25$.

The Universal Set

- A set U that includes all of the elements under consideration in a particular discussion.
- Depends on the context.
- Examples: The set of Latin letters, the set of natural numbers, the set of points on a line.

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Equality of Sets (1/3)

- Two sets A and B are equal, denoted $A = B$, if they have the same elements.
- Otherwise, $A \neq B$.
- Example:
 - The set A of odd positive integers is not equal to the set B of prime numbers.
 - The set of odd integers between 4 and 8 is equal to the set of prime numbers between 4 and 8.

Equality of Sets (2/3)

- Example: The sets $A = \{1, 2, 3, \dots, n, \dots\}$ and $B = \{2, 4, 6, \dots, 2n, \dots\}$ are equivalent because we can establish the correspondence $n \mapsto 2n$ ($n = 1, 2, 3, \dots$).
 - It is noteworthy that B is a proper subset of A .
- Thus, for infinite sets, a proper subset of X can be equivalent in cardinality to X itself.
- In that case: if A is equivalent to a subset B' of B , but A is not equivalent to B , then $|A| < |B|$.

Equality of Sets (3/3)

- Example: Prove that the set of points in $(0,1)$ is equivalent in cardinality to the set of points in (a,b) , where (a, b) are arbitrary real numbers.
- Hint: Consider the bijection $f(x) = a \cdot x + b \cdot (1 - x)$ with $x \in (0,1)$.

Theorem 1. Cantor–Bernstein Theorem. If A is equivalent to a subset B' of B , and B is equivalent to a subset A' of A , then A and B are equivalent.

- This shows that between two sets A and B , there are four possible relationships:
 $|A| = |B|$, $|A| > |B|$, $|A| < |B|$, or A and B are incomparable.

Countable sets

- A set is **countable** (finite or infinite) if we can enumerate its elements in the form:

$$a_1, a_2, \dots$$

- Among infinite sets, a countable set has the smallest cardinality. Any set equivalent to a countable set is also countable.

Theorem 2. Every infinite set has a countable subset.

A subset of a countable set is either finite or countably infinite.

The union of finite or countably infinite sets is countable.

Adding a finite set or a countably infinite set to an infinite set does not change its cardinality.

The set of real numbers is uncountable (its cardinality is the continuum).

Subsets

- A is a subset of B if every element of A is an element of B.
 - Notation: $A \subseteq B$
- For each set A, $A \subseteq A$.
- For each set B, $\emptyset \subseteq B$.
- A is proper subset of B if $A \subseteq B$ and $A \neq B$.
 - Notation: $A \subset B$.

Subsets

- Given a set X , the set of all subsets of X (also called the **power set** of X) is denoted by $P(X)$.
- In other words, $P(X)$ is a set whose elements are the subsets of X .

- Example: If $X = \{0,1,2\}$, then

$$P(X) = \{\emptyset, \{0\}, \{1\}, \{2\}, \{0,1\}, \{0,2\}, \{1,2\}, \{0,1,2\}\}.$$

Questions:

- If $X \subseteq Y$ then $P(X) \subseteq P(Y)$. Is the converse true?
- If X has n elements ($n \in \mathbb{N}$), then $P(X)$ has 2^n elements.

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Mapping

- A rule f that assigns each element $x \in X$ to a uniquely determined element $y \in Y$ is called a mapping (function). It is denoted by $y = f(x)$.

- Let A, B be subsets of X, Y .

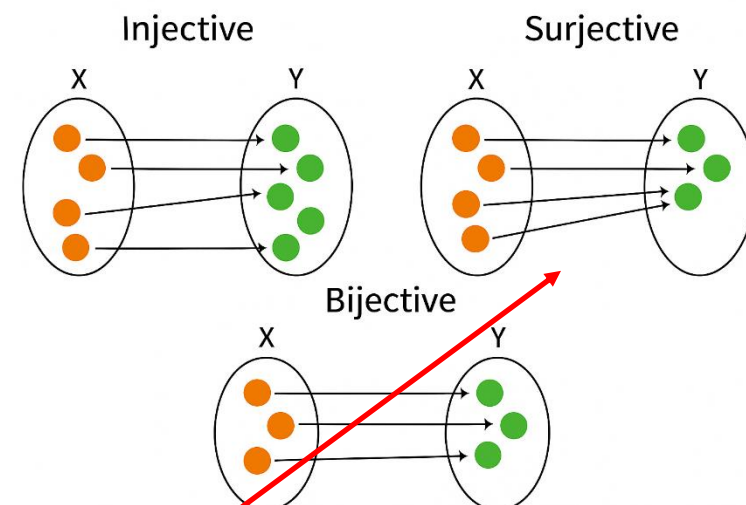
- Then $f(A)$ is the image of A , and $f^{-1}(B)$ is the preimage of B .

- Injective (one-to-one): if $x \neq x'$ then $f(x) \neq f(x')$.

- Surjective (onto): for every $y \in Y$, there exists $x \in X$ such that $f(x) = y$.

- Bijjective (one-to-one correspondence): both injective and surjective.

- The composition of two mappings $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ is $f \circ g: X \rightarrow Z$.



Outline

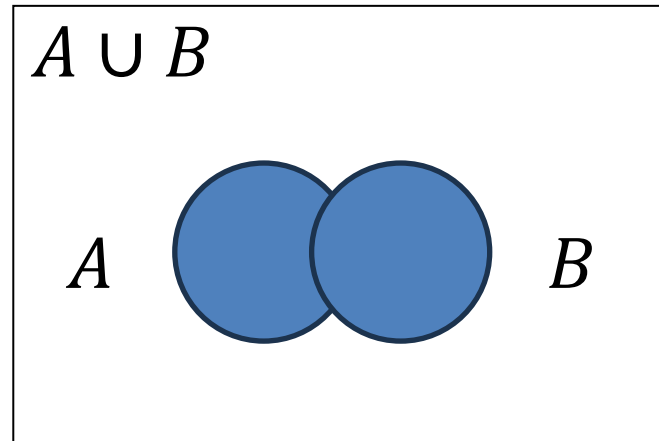
1. Basic definitions
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Unions

- The union of two sets A and B is

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$

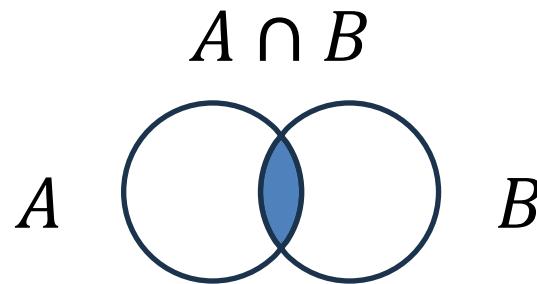
- The word “or” is inclusive.



Intersections

- The intersection of A and B is

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$



- Example: Let A be the set of even positive integers and B the set of prime positive integers. Then $A \cap B = \{2\}$.
- Definition: A and B are disjoint if $A \cap B = \emptyset$.

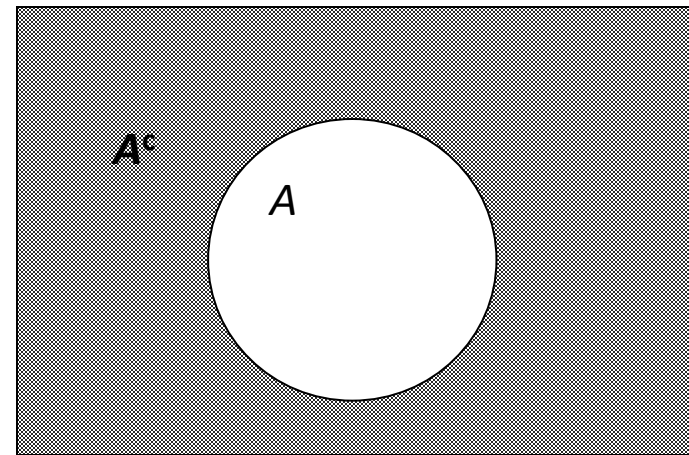
Complements

- If A is a subset of the universal set U , then the complement of A is the set

$$A^c = \{x \in U \mid x \notin A\}.$$

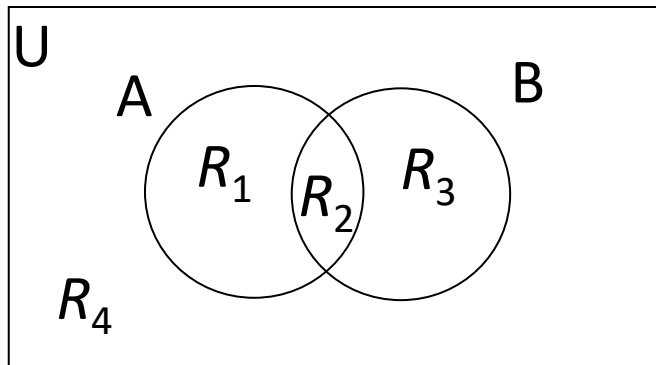
- Note:

- $A \cap A^c = \emptyset$
- $A \cup A^c = U$



The shaded region represents the complement of the set A

Sets Formed by Two Sets



$$R_1 = A \cap B^c$$

$$R_2 = A \cap B$$

$$R_3 = A^c \cap B$$

$$R_4 = A^c \cap B^c$$

Two Basic Counting Rules

If A and B are finite sets,

1. $|A \cup B| = |A| + |B| - |A \cap B|.$

2. $|A \cap B^c| = |A| - |A \cap B|.$

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Three basic counting principles: The Addition Principle

The Addition Principle. If there r_1 different objects in the first set, r_2 different objects in the second set, . . . , and r_m different objects in the m -th set, and *if the different sets are disjoint*, then the number of ways to select an object from one of the m sets is $r_1 + r_2 + \cdots + r_m$.

Example 1

Students choose seats at random in **3 different classes**. There are **23**, **15**, and **19** seats in the classes, respectively. How many ways can a student choose their seats?

The Addition Principle

Remark 1. Calculate the total number of possible outcomes when combining the sets. If $A_1, A_2, \dots, A_m$ are the sets representing the seats in each class, then the total number of outcomes is the sum of the individual sets:

$$|A_1 \cup A_2 \cup \dots \cup A_m| = |A_1| + |A_2| + \dots + |A_m|.$$

Outline

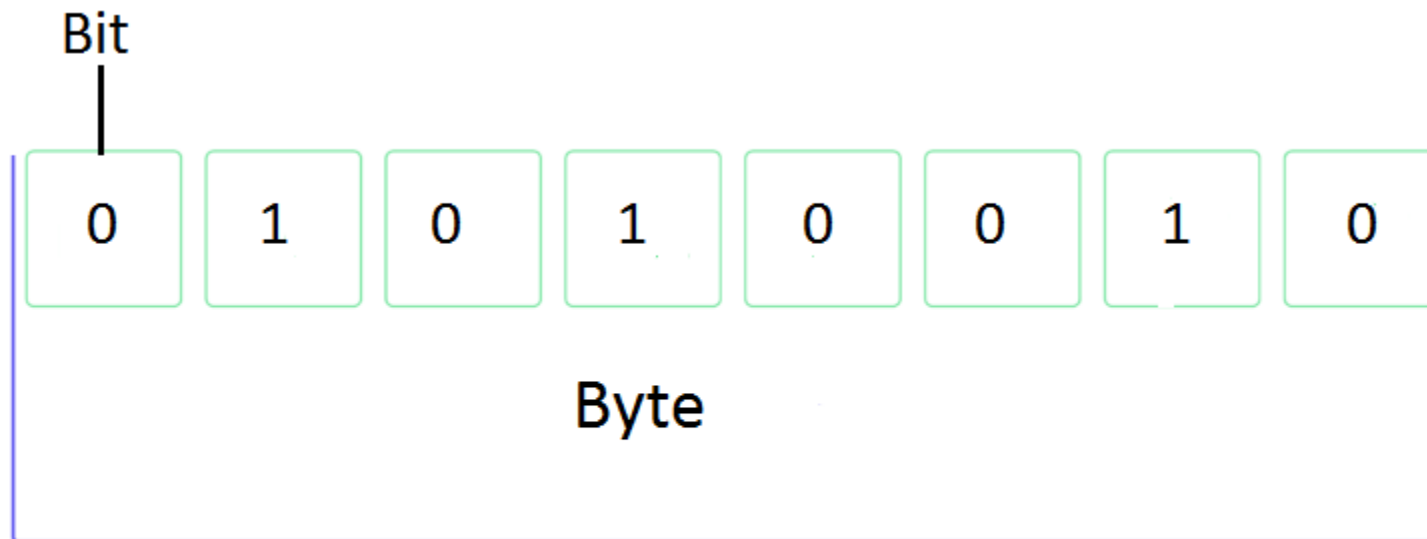
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Three basic counting principles: The Multiplication Principle

The Multiplication Principle. Suppose a procedure can be broken into m successive (ordered) stages, with r_1 different outcomes in the first stage, r_2 different outcomes in the second stage, . . . , and r_m different outcomes in the m -th stage. If the number of outcomes at each stage is independent of the choices in previous stages and *if the composite outcomes are all distinct*, then the total procedure has $r_1 \times r_2 \times \cdots \times r_m$ different composite outcomes.

Example 2

How many ways can an 8-bit sequence be formed?



8 bits = 1 Byte

Example 2

How many ways can an 8-bit sequence be formed?

Solution. There are 2 choices per bit: 0 or 1. To form a sequence of 8 bits, according to the principle of multiplication, the total number of ways is

$$2^8 = 256.$$

The Multiplication Principle

Remark 2. Calculate the number of possible outcomes when combining the sets. If $A_1, A_2, \dots, A_m$ are the sets representing the possibilities of each bit, then the total number of sequence outcomes is the product of the individual sets:

$$|A_1 \times A_2 \times \cdots \times A_m| = |A_1| \times |A_2| \times \cdots \times |A_m|.$$

Example 3

Set A has 6 elements and set B has 10 elements. How many:

- a) Mappings from A to B ?
- b) Injective function from A to B ?

Example 3

Solution.

a) For each element x in A , we have 10 ways to choose the image due to B has 10 elements. To construct a mapping from A to B , we take in turns image of 6 elements of A . Following the multiplication principle, we have $10 \times 10 \times \cdots \times 10 = 10^6$ mapping from A to B .

6 times

Example 3

Solution. Injective function from A to B ?

b) Suppose that $A = \{x_1, x_2, \dots, x_6\}$. The problem is divided into 6 steps:

- Step 1: To pick an image of x_1 , we have 10 ways.
- Step 2: To pick an image of x_2 , we have $10 - 1 = 9$ ways.
- ...
- Step 6: To pick an image of x_6 , we have $10 - 5 = 5$ ways.

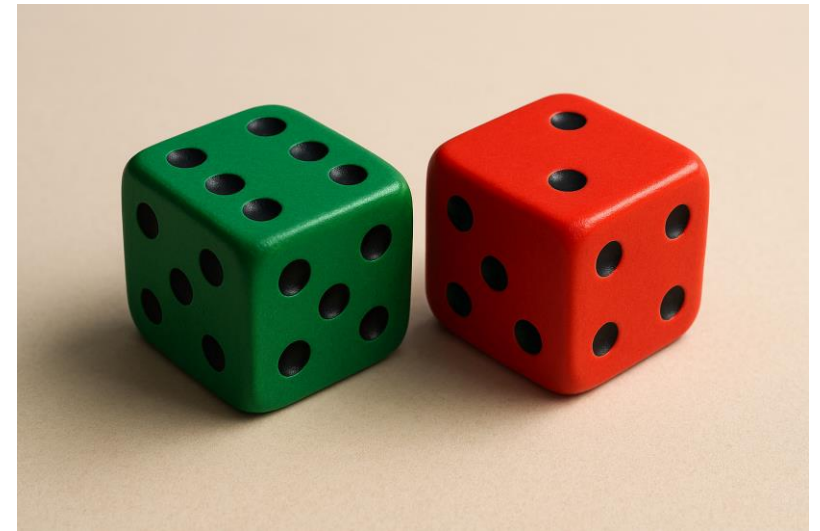
So that, the number of injection mapping from A to B : $\prod_{i=5}^{10} i = 151200$.

Example 4: Rolling Dice

Two die are rolled, one green and one red.

Each dice has faces numbered 1 through 6.

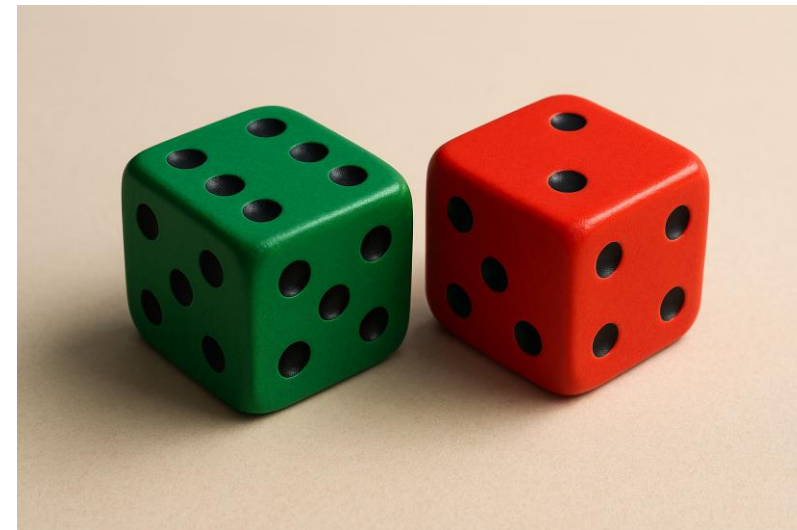
- a) How many different outcomes of this procedure are there?
- b) What is the probability that there are no doubles (not the same value on both dice)?



Example 4: Rolling Dice

- a) How many different outcomes of this procedure are there?

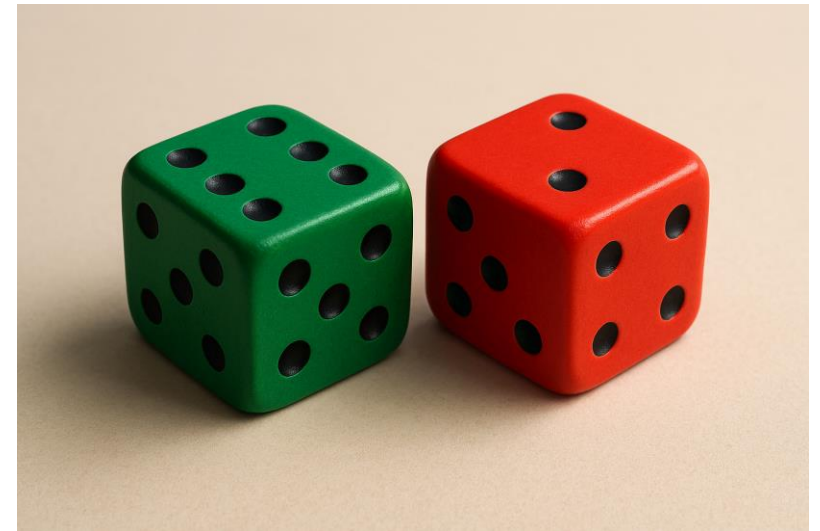
Solution. $6 \times 6 = 36$ ways.



Example 4: Rolling Dice

b) What is the probability that there are no doubles (not the same value on both dice)?

Solution. The value of the green die is different from the value of the red die. So, we have $6 \times 5 = 30$ outcomes with no doubles. Thus, the probability of no doubles is $\frac{30}{36} = \frac{5}{6}$.



Example 5: Arranging Books

There are **five different Spanish books**, **six different French books**, and **eight different Transylvanian books**. How many ways are there to pick an (unordered) pair of two books **not both** in the same language?



Example 5: Arranging Books

Solution. There are three cases:

- First, two books consist of a Spanish and a French book: 5×6 .
- Second, a Spanish and a Transylvanian: 5×8 .
- Third, a French and a Transylvanian book: 6×8 .

Thus, we have:

$$5 \times 6 + 5 \times 8 + 6 \times 8 = 30 + 40 + 48 = 118.$$

Example 6: Sequences of Letters

How many ways are there to form a three-letter sequence using the letters a, b, c, d, e, f

- i. with repetition of letters allowed? 216
- ii. without repetition of any letter? 120
- iii. without repetition and containing the letter e ? 20
- iv. with repetition and containing e ? 91

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Dirichlet's principle

Some interesting facts:

- Among 367 people, there are at least 2 people with the same birthday.
- There are 20 birds trapped in 7 cages. Prove that there exists a cage with at least 3 birds.

Dirichlet's principle

Definition 1. The ceiling value of x , denoted as $\lceil x \rceil$, is the smallest integer greater than or equal to x .

Example: $\lceil 2.1 \rceil = 3$; $\lceil 1.9 \rceil = 2$; $\lceil 4 \rceil = 4$; $\lceil -1.1 \rceil = -1$; $\lceil -2.9 \rceil = -2$.

Dirichlet's principle

Dirichlet's principle. If n objects are placed into k boxes, there is at least one box containing at least $\left\lceil \frac{n}{k} \right\rceil$ objects.

Dirichlet's principle Proof

Proof.

Suppose that every box contains less than $\left\lceil \frac{n}{k} \right\rceil$. Then, the total items have to less than or equal

$$k \left(\left\lceil \frac{n}{k} \right\rceil - 1 \right) < k \times \frac{n}{k} = n.$$

This is a contradiction because we have n items.

Example 7

How many students must there be in a class to have at *least 6 students with the same learning level*, knowing that there are **5 types** of learning levels: A, B, C, D, and E?



Example 7

How many students must there be in a class to have *at least 6 students with the same learning level*, knowing that there are **5 types** of learning levels: A, B, C, D, and E?

Solution. Let N is the number of students in class. Following the Dirichlet's Principle, we have: $\left\lceil \frac{N}{5} \right\rceil \geq 6$. Then, $N > 25$.

Thus, we choose $N = 26$, i.e, this class has at least 26 students.

Example 8

Show that among any 10 natural numbers we can choose two numbers whose difference is divisible by 9.

Example 9

Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and A is the subset of X which has 6 elements. Show that, in A should have two element which sum of them is exactly 10.

Example 10

In a meeting room with n people, prove that there are always 2 people who have the same number of acquaintances among the meeting attendees.

Solution. Left as an exercise to the students.

Example 11

Selecting $n + 1$ natural numbers from 1 to $2n$, show that there always exist 2 co-prime integers.

Solution. Left as an exercise to the students.

Homework (1/2)

1. How many odd numbers from 100 to 999 have distinct digits?
2. From the letters a, b, c, d, e, f, how many ways can we form a string of 3 characters:
 - a) Freely?
 - b) With all characters distinct?
 - c) With all characters distinct and the character f must appear?
3. A train with 5 cars stops at a station to pick up 3 passengers. In how many ways can these 3 passengers get on the train if:
 - a) Each passenger can board any car?
 - b) Each passenger boards a different car?
 - c) All 3 passengers board the same car?

Homework (2/2)

4. In a month with 30 days, a volleyball team plays at least one match every day but no more than 45 matches in total. Prove that there exists a consecutive period of days during which the team plays exactly 14 matches.

5. Suppose that in a group of 6 people, each pair of people are either friends or enemies. Prove that in the group, there are either three people who are all mutual friends or three people who are all mutual enemies.

Q & A